



2004

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Biology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- Write your Centre Number and Student Number at the top of pages 9, 13, 15 and 19

Total marks – 100

Section I Pages 2–21

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–29
- Allow about 1 hour and 45 minutes for this part

Section II Pages 23–29

25 marks

- Attempt ONE question from Questions 30–34
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

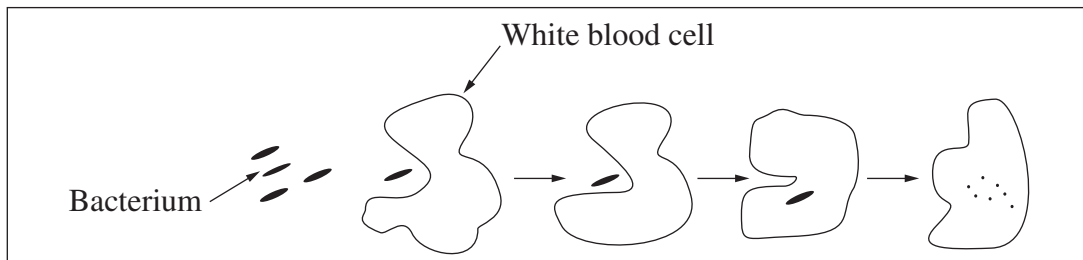
A ☒ B ☒ C ☐ D ☐
 correct
 ↙

- 1 Which of the following statements can be used to describe a pathogen?
- (A) They are all viral.
 - (B) They are all infectious.
 - (C) They are all microscopic.
 - (D) They are all macroparasites.
- 2 Which of the following helps prevent the entry of pathogens into humans?
- (A) Cilia
 - (B) Antigens
 - (C) B lymphocytes
 - (D) Inflammation response
- 3 Recently, hospitals and medical practitioners have warned the community about the spread of severe acute respiratory syndrome (SARS). People were experiencing:
- high temperatures
 - body aches
 - pains similar to that of the flu.

How would you classify these descriptions?

- (A) Controls
 - (B) Symptoms
 - (C) Warnings
 - (D) Methods of transmission
- 4 What is the name of the scientist who identified the role of microbes in decay?
- (A) Macfarlane Burnet
 - (B) Robert Koch
 - (C) Louis Pasteur
 - (D) Ronald Ross

- 5 The diagram below illustrates one process that occurs as part of an immune response.



What process does the diagram illustrate?

- (A) Cytokinesis
(B) Inflammation
(C) Osmosis
(D) Phagocytosis
- 6 Which adaptation assists temperature regulation in plants?
- (A) Large leaves for cooling
(B) Increased production of seeds
(C) Movement of glucose to roots
(D) Evaporation of water from stomates
- 7 Blood travels around the body through arteries, veins and capillaries. Which of the following correctly describes the structure or function of each of these blood vessels?

	<i>Artery</i>	<i>Vein</i>	<i>Capillary</i>
(A)	Always carries oxygenated blood	Always carries deoxygenated blood	Always carries waste products
(B)	Thick muscular walls	Thin walls with valves	Walls one cell thick
(C)	Takes blood to the heart	Thick muscular walls	Takes blood to and from the heart
(D)	Carries blood under high pressure	Carries blood under variable pressure	Thin walls with valves

- 8 Which statement best describes a function of nephrons?
- (A) Filter waste products from the blood
 - (B) Remove carbon dioxide from the blood
 - (C) Collect urine from the blood
 - (D) Remove the waste products directly from the cells in the blood
- 9 Which statement defines homeostasis in multicellular organisms?
- (A) Homeostasis is the process by which cells maintain their internal environment.
 - (B) Homeostasis is the maintenance of the internal and external environment of the organism.
 - (C) Homeostasis is the process by which animals and plants maintain their body temperature.
 - (D) Homeostasis is the maintenance of a constant internal environment of the organism.
- 10 Which of the following sequences most correctly represents the results of a reaction involving an enzyme?

KEY

E	Enzyme
P_1	Product ₁
P_2	Product ₂
S	Substrate

- (A) $P_1 + S \rightarrow P_2 + E$
 - (B) $S + E \rightarrow P_1 + P_2$
 - (C) $P_1 + P_2 + E \rightarrow S + E$
 - (D) $S + E \rightarrow P_1 + P_2 + E$
- 11 Which of the following is the correct base-pairing in DNA?
- (A) G—C, A—T
 - (B) T—U, A—G
 - (C) G—T, A—C
 - (D) G—C, A—U

- 12** Reproductive technologies focus on the transfer of genetic information.

Which process only involves the transfer of the nucleus?

- (A) Cloning
- (B) Transgenesis
- (C) Artificial pollination
- (D) Artificial insemination

- 13** In rabbits, black (B) coat colour is dominant over white (b) coat colour.

Which Punnet square correctly represents a cross between a rabbit heterozygous for coat colour and a white rabbit?

(A)

	B	b
B	BB	Bb
b	Bb	bb

(B)

	B	B
b	Bb	Bb
b	Bb	Bb

(C)

	B	b
b	Bb	bb
b	Bb	bb

(D)

	B	b
b	Bb	Bb
b	Bb	Bb

- 14** Why is accurate replication of DNA important?

- (A) It leads to cell differentiation.
- (B) It maintains genetic information.
- (C) It allows for evolution of the species.
- (D) It enables cells to modify their proteins.

- 15** 'Generally, gametes produced by an organism will not be identical.'

Which of the following does NOT influence this genetic variability of gamete formation?

- (A) Mutation
- (B) Sex linkage
- (C) Independent assortment of alleles
- (D) Crossing over in homologous chromosomes

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Centre Number

Section I (continued)

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Student Number

Part B – 60 marks

Attempt Questions 16–29

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

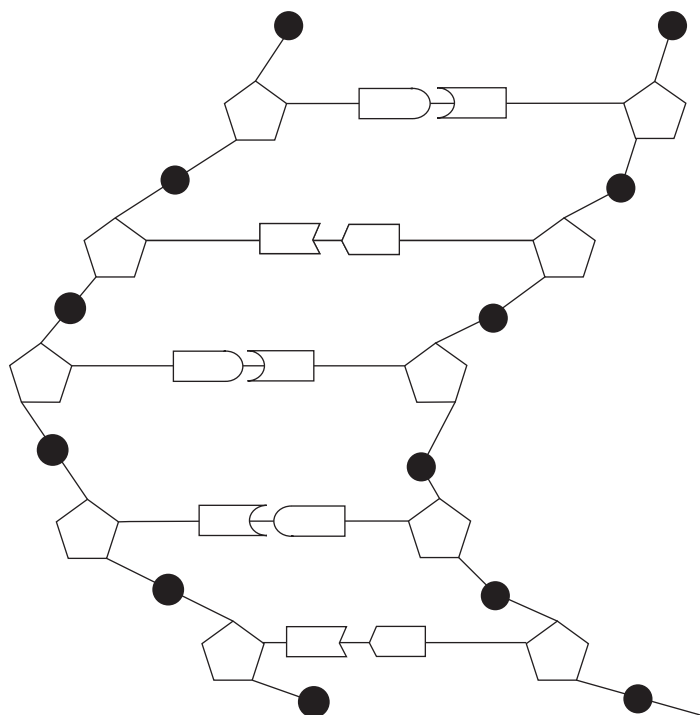
Marks

Question 16 (4 marks)

- (a) On the diagram, clearly identify ONE nucleotide by placing a box around it.

1

Structure of portion of a DNA molecule



Senior Biology, RJ King and FM Sullivan, Longman Australia/Pearson Education Australia. Reproduced with permission.

- (b) Outline the main steps of DNA replication.

3

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Question 17 (7 marks)

A plant breeder replicated one of Mendel's pea plant breeding experiments. The results of the cross between green (G) pods and yellow (g) pods are shown in the table.

<i>Generation</i>	<i>Pod colour</i>	
	<i>Green (G)</i>	<i>Yellow (g)</i>
F_1	632	0
F_2	1560	482

- (a) List all possible genotypes of the F_2 generation. **1**

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- (b) Using your understanding of current genetic thinking, interpret the results of the F_2 cross depicted above. **3**

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- (c) Describe the features of TWO experimental techniques used by Mendel that led to his success. **3**

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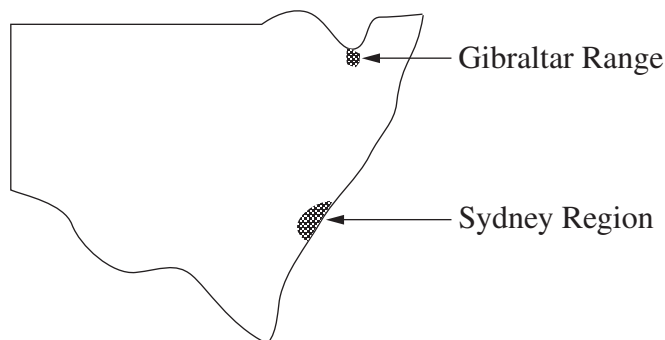
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Question 18 (6 marks)

- (a) A plant species found in the area immediately around Sydney has also been found in a small area in the Gibraltar Range in the far north of NSW. **3**



Map of NSW

Predict what might happen to the TWO populations over the next 5 million years, in terms of Darwin/Wallace's theory of evolution.

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- (b) Justify the use of vertebrate forelimbs as evidence to support the theory of evolution. **3**

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Centre Number

Section I – Part B (continued)

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Student Number

Question 19 (4 marks)

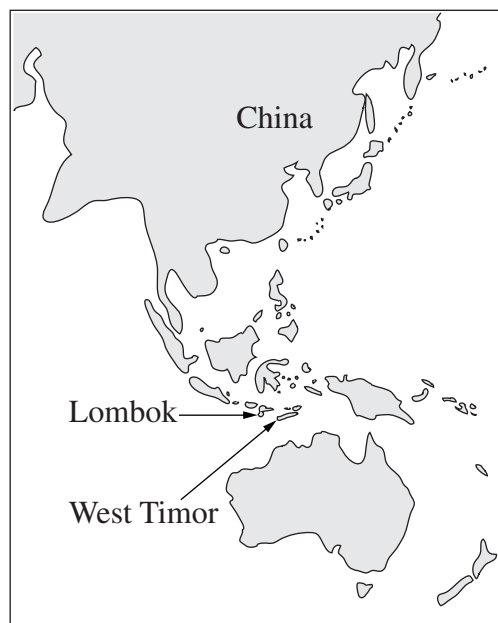
Marks

Further avian influenza outbreaks in Indonesia

Avian influenza has been found closer to Australian shores, with more outbreaks in Indonesia.

The spread of the disease has slowed across most Asian countries, but in Indonesia it has spread as far south as Lombok and West Timor.

The Australian Chicken Growers Council says the new outbreaks don't mean Australia's chances of contracting the disease are any greater, but high quarantine measures remain in place.



Using avian influenza as an example, evaluate the effectiveness of quarantine measures in Australia.

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Question 20 (3 marks)

- (a) Identify ONE type of *T* lymphocyte. **1**

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- (b) Distinguish between the functions of *B* cells and *T* cells. **2**

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Question 21 (3 marks)

The Murray-Darling Basin is the natural habitat of the Murray Cod, a native Australian freshwater fish. **3**

Explain the implications of increased water salinity for the survival of the Murray Cod.

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Question 22 (3 marks)

Public health programs, pesticides and genetic engineering are used to control and/or prevent disease. **3**

Using an example, explain how ONE of these strategies has been used to control or prevent disease within the community.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 23** (3 marks)

Current theories describe differences in the movement of substances through plants.

3

Compare the movement of substances in xylem and phloem tissues.

<i>Feature</i>	<i>Xylem</i>	<i>Phloem</i>
Material transported		
Process of movement		
Name of relevant current theory		

Question 24 (3 marks)

Justify continued research into the development of artificial blood.

3

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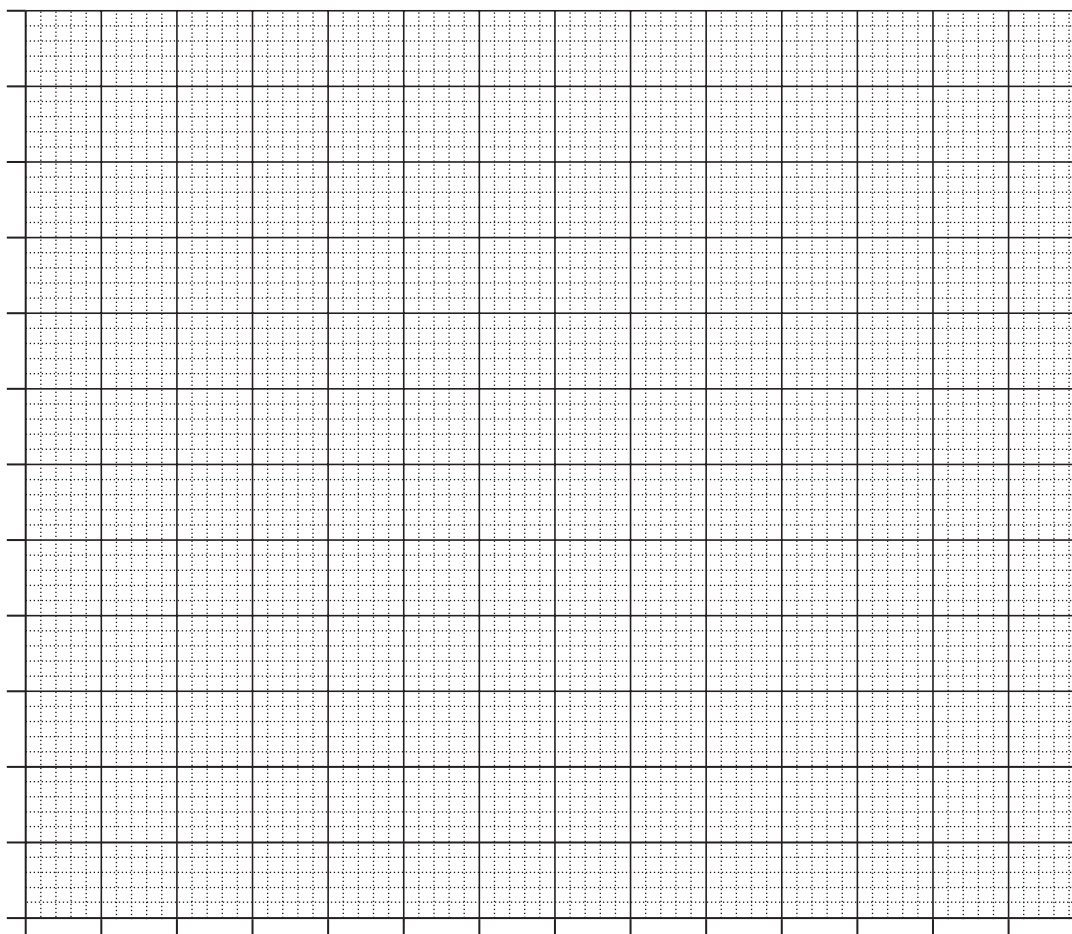
Question 25 (6 marks)

(a)

Death due to lung cancer in males and females in Australia**5**

<i>Time</i>	<i>Death rate per 100 000 population</i>	
	<i>Males</i>	<i>Females</i>
1994	59	19
1995	56	19
1996	55	19
1997	51	19
1998	52	18
1999	50	19
2000	48	19
2001	47	20

Draw the most appropriate graph for the information, on the grid provided.

Death due to lung cancer

Question 25 continues on page 17

Question 25 (continued)

- (b) Suggest additional data that needs to be gathered before a relationship between smoking and lung cancer can be inferred. **1**

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Question 26 (3 marks)

You undertook a first-hand investigation to examine plant shoots and leaves for evidence of pathogens and insect pests. **3**

Using the information you gathered, complete the table.

	<i>Plant shoot or plant leaf</i>
Evidence of pathogen	
Evidence of insect pest	
Equipment or procedure used for observation	

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Centre Number

Section I – Part B (continued)

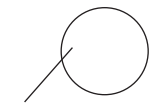
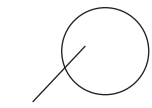
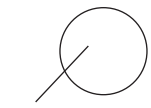
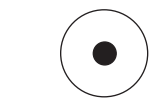
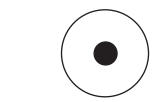
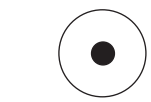
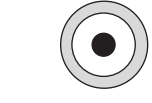
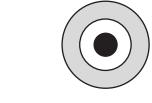
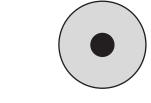
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Student Number

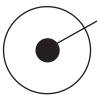
Question 27 (4 marks)

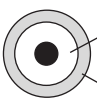
Marks

To study the effect of an antibiotic on three strains of bacteria (*A*, *B*, *C*), agar plates were set up as shown.

	Agar plate 1	Agar plate 2	Agar plate 3
Method			
	Surface covered with bacterium <i>A</i>	Surface covered with bacterium <i>B</i>	Surface covered with bacterium <i>C</i>
			
	Incubation 37°C, 48 hours		
Result			

KEY

 Disc of the same antibiotic *X* placed on each plate

 Clear zone no bacteria

 Bacterial growth

The plates were incubated at 37°C for 48 hours. The diagrams of plates 1, 2 and 3 depict the results.

- (a) Write a conclusion to the experiment.

1

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- (b) Identify ONE safe work practice used to minimise risks associated with handling or identifying microbes.

1

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Question 27 continues on page 20

Question 27 (continued)

- (c) In humans, bacterial infections are often treated with antibiotics. Explain why the complete course of antibiotics should be taken, even if the symptoms of infections have disappeared. **2**

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Question 28 (3 marks)

‘All mutations are harmful.’ **3**

Discuss this statement.

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8

	<i>Environmental conditions where bacteria were collected</i>	
	<i>Temperature (°C)</i>	<i>pH</i>
Bacteria <i>X</i>	50–80	5.5–7.5
Bacteria <i>E</i>	37–55	6.5–8.5

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Biology

Section II

25 marks

Attempt ONE question from Questions 30–34

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

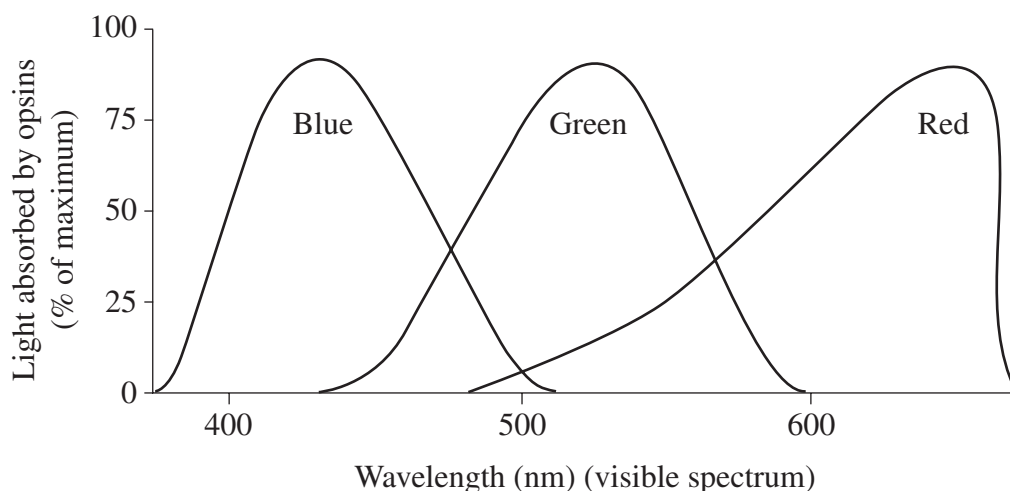
	Pages
Question 30 Communication	24
Question 31 Biotechnology	25
Question 32 Genetics: The Code Broken?	26
Question 33 The Human Story	27–28
Question 34 Biochemistry	29

Question 30 — Communication (25 marks)

- (a) (i) Name the part of the brain involved in perception and interpretation of light and sound. **1**
- (ii) Describe the stimulus-response pathway that is used to respond to stimuli from the external environment. **3**
- (b) During the study of Communication, you undertook an investigation to identify the relationship between wavelength, frequency and pitch of sound.
- (i) Provide an outline of the method followed in this investigation. **4**
- (ii) Explain how this investigation increased your understanding of sound. **2**
- (c) A focus of the option Communication was the technologies developed to overcome difficulties associated with the detection of, transmission of and response to stimuli in the environment. **7**

Discuss the impact that these technologies have had on society.

- (d) The graph depicts the relationship between light absorption by colour sensitive pigments (opsins) and wavelength of light.



Biology Options Communication, Glenda Chidrawi and Marilyn Mercer, 2003, McGraw- Hill Australia Pty Ltd, Sydney.

- (i) Explain why the brain is able to interpret a variety of colours. **2**
- (ii) With reference to light absorption of opsins, explain how a person can be colour blind. **2**
- (iii) Describe the use of colour for communication in animals other than humans, and relate this to the occurrence of colour vision in animals. **4**

Question 31 — Biotechnology (25 marks)

- (a) (i) Name an ancient Australian Aboriginal use of biotechnology. **1**
- (ii) Describe how the breeding of animals with desired characteristics may be considered as early biotechnology. **3**
- (b) During the study of Biotechnology, you undertook an investigation to extract DNA from a suitable source.
- (i) Provide an outline of the method followed in this investigation. **4**
- (ii) Outline uses of extracted DNA in biotechnology. **2**
- (c) A focus of the option Biotechnology was an understanding of biotechnological processes and the application of these processes for assisting humans. **7**

Discuss, using named examples, the impact of the application of biotechnology on society.

(d)



Fluorescent fish go on sale

Despite objections from food safety and conservation groups, zebra fish genetically modified to fluoresce red went on sale in several pet stores in Florida last week. 'They're selling really well,' says Steven Feinberg of the Pet Supermarket chain. Each GloFish costs \$4.99, ten times as much as a normal zebra fish, he says.

For this genetically modified fish to be produced, a number of processes would have been used. Gene splicing may have been one of them.

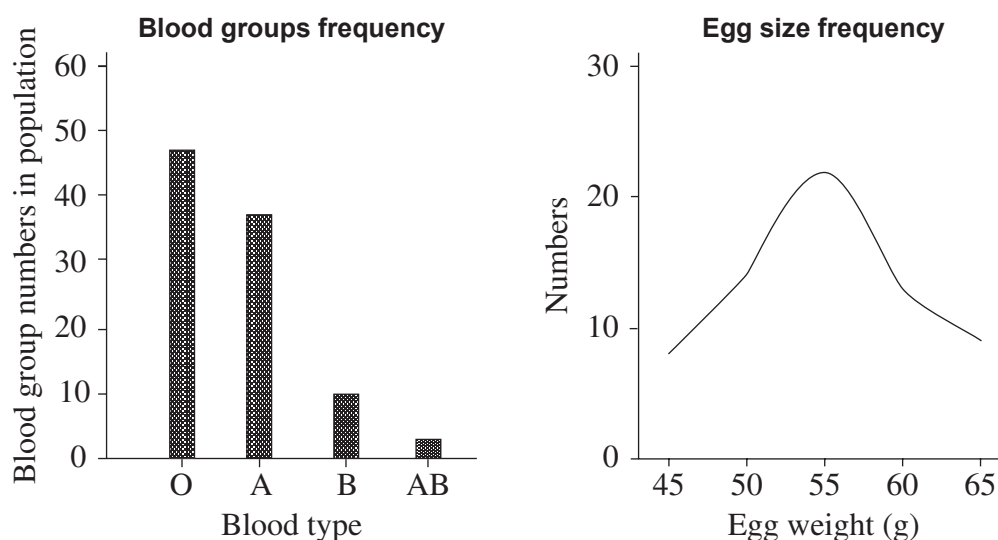
- (i) Outline the process of gene splicing to produce recombinant DNA. **2**
- (ii) Fluorescent zebra fish are able to mate with non-fluorescent fish to produce viable offspring. Propose an impact on the environment if the fluorescent fish are accidentally released. **2**
- (iii) Explain why some groups in society may have different views about the use of DNA technology to produce zebra fish that fluoresce. **4**

Question 32 — Genetics: The Code Broken? (25 marks)

- (a) (i) Name the nitrogen base unique to mRNA. **1**
- (ii) Outline how the information contained in DNA is used to produce a polypeptide. **3**
- (b) During the study of Genetics: The Code Broken?, you undertook an investigation to model linkage.
- (i) Provide an outline of the method followed when modelling linkage. **4**
- (ii) Explain how this investigation increased your understanding of linkage. **2**
- (c) A focus of the option Genetics: The Code Broken? included genetic changes that occur as a result of human intervention. **7**

Discuss the impact that genetic change due to human intervention has had on society.

- (d) In two separate experiments, blood groups in humans and egg size in chickens were investigated. The following data were collected and the results graphed.



- (i) Identify the graph that represents polygenic inheritance. **1**
- (ii) Discuss the differences between the graphs, supporting your answer with a description of polygenic and multiple allele inheritance. **4**
- (iii) Using an example, outline the use of highly variable genes for DNA fingerprinting of forensic samples. **3**

Question 33 — The Human Story (25 marks)

- | | | | |
|-----|---|---|----------|
| (a) | (i) | Name the GENUS to which modern humans belong. | 1 |
| | (ii) | Describe THREE primate characteristics of prosimians. | 3 |
| (b) | During the study of The Human Story, you undertook an investigation to model DNA-DNA hybridisation. | | |
| | (i) | Provide an outline of the method followed when modelling DNA-DNA hybridisation. | 4 |
| | (ii) | Explain how this investigation increased your understanding of the use of DNA-DNA hybridisation in possible evolutionary relationships. | 2 |
| (c) | A focus of the option The Human Story was the use of fossil evidence to support various theories about human evolution. | | 7 |

Discuss the impact that modern technologies have on research in this area.

Question 33 continues on page 28

Question 33 (continued)

- (d) The three sets of diagrams below illustrate some anatomical differences between apes and humans.



- | | |
|--|---|
| (i) Based on the data above, construct a table that describes the anatomical differences between an ape and a human. | 4 |
| (ii) How does the interpretation of the differences in TWO of these anatomical features provide evidence to support human evolution? | 4 |

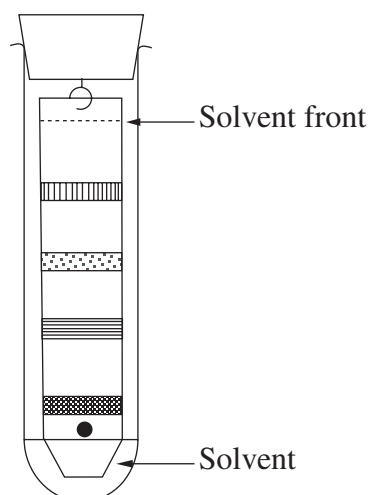
End of Question 33

Question 34 — Biochemistry (25 marks)

- (a) (i) Name the organelle where photosynthesis occurs. **1**
- (ii) Describe the overall process of photosynthesis. **3**
- (b) During the study of Biochemistry, you undertook an investigation to determine the effect of light intensity and temperature on gas production in a pond weed.
- (i) Provide an outline of the method followed in this investigation. **4**
- (ii) Explain how this investigation increased your understanding of photosynthesis. **2**
- (c) A focus of the option Biochemistry includes the study of reactions within living organisms. **7**

Discuss the impact of new technologies on biological research.

- (d) Tswett invented a method for the separation of leaf pigments. The diagram represents possible results of Tswett's work.



Senior Biology, RJ King and FM Sullivan, Longman Australia 1995/Pearson Education Australia. Reproduced with permission.

- (i) Name the method invented, and outline how Tswett would have interpreted the results. **2**
- (ii) Describe the role of chlorophyll in the light reaction. **2**
- (iii) Referring to a named scientist, explain the role of radioactive isotopes in determining our current understanding of photosynthesis. **4**

End of paper

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**2004 HSC Notes from
the Marking Centre
Biology**

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2004 HSC NOTES FROM THE MARKING CENTRE

BIOLOGY

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Biology. It provides comments with regard to responses to the 2004 Higher School Certificate Examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section and each question.

It is essential for this document to be read in conjunction with the relevant syllabus, the 2004 Higher School Certificate Examination, the Marking Guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Biology.

General Comments

In 2004, approximately 13 000 candidates attempted the Biology examination. The most popular option in 2004 was *Communication* followed by *Genetics – The Code Broken?* and the *Human Story*. The *Biochemistry* option was selected by few candidates in 2004.

Teachers and candidates should be aware that examiners may ask questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately. There was evidence that candidates who spent some time planning a concise response to the specific question being asked tended to achieve better marks. Candidates should be guided by the key word and the mark allocation for the depth of treatment required in a response.

Overall, the candidate's responses were appropriate and indicated a level of understanding of Biology concepts, appropriate for HSC candidates. Candidates need to be reminded that the answer space allocated is a guide to the maximum length of response required. Similarly, the key word used in the question gives an indication of the depth of the required response. The option question is divided into a number of parts: candidates should clearly label each part of the question when writing in their answer booklets. A small number of candidates responded to more than one question in this section and there was some evidence of candidates attempting options that they had not studied. All these candidates tended to score low marks.

Section I – Core

Part A – Multiple choice

Question	Correct Response
1	B
2	A
3	B
4	C
5	D
6	D
7	B
8	A

Question	Correct Response
9	D
10	D
11	A
12	A
13	C
14	B
15	B

Part B

Specific Comments

Question 16

- (a) A significant number of candidates were unable to identify a nucleotide.
- (b) In the better responses candidates outlined three stages of DNA replication in their correct sequence. A significant number of candidates confused the process of DNA replication with the process of protein synthesis.

Question 17

- (a) Most candidates were able to correctly list the three possible genotypes.
- (b) In the better responses, candidates were able to interpret the results of the cross in terms of the genotypes of the parents, the dominance/recessiveness of each characteristic and the ratios shown.
- (c) In the better responses candidates were able to provide the features of two experimental techniques used by Mendel and clearly show how they led to his success.

Question 18

- (a) In the better responses candidates made valid suggestions about what may happen for both populations in terms of the Darwin/Wallace Theory of Evolution. A significant number of candidates confused the distribution of organisms with their abundance.
- (b) In the better responses, candidates provided examples of vertebrate forelimbs, demonstrated an understanding of divergent evolution and related common structures to common ancestry as evidence supporting the theory of evolution.

Question 19

In the better responses, candidates described at least two quarantine measures, explained how they assist in protecting Australia and made judgements about the effectiveness of the measures described. A significant number of candidates simply described quarantine measures without making judgements about their effectiveness.

Question 20

- (a) Most candidates were able to correctly identify one type of T lymphocyte.
- (b) A significant number of candidates had difficulty in providing distinct or different features of the functions of the two types of cells, and instead wrote about where each type of cell was made.

Question 21

In the better responses, candidates clearly linked the increased water salinity to survival of the Murray Cod by relating it to the water balance mechanisms of the fish.

Question 22

Most candidates were able to explain correctly how one of the three strategies is used to control and/or prevent disease. Some candidates simply stated examples of one of the strategies.

Question 23

Most candidates correctly identified the material transported in xylem and phloem, along with the process of movement. A significant number of candidates were unable to name or describe the relevant current theories to explain the movement of substances.

Question 24

This question was generally well answered, with most candidates able to support arguments for continued research into the development of artificial blood.

Question 25

This question assessed skills in both choosing the most appropriate graph and drawing it correctly, and was generally well answered.

- (a) In the better responses, candidates correctly chose to draw a histogram. A significant number of candidates chose to draw a line graph (polygon), but correctly transposed data, labelled axes, chose an appropriate scale and included a key.
- (b) Most candidates correctly suggested additional relevant data that needs to be collected.

Question 26

This question was generally well answered. In the better responses, candidates clearly understood the difference between a pathogen and an insect pest. Most candidates were able to give a piece of equipment or procedure used in the investigation that was relevant to the evidence provided.

Question 27

- (a) This part of the question was generally well answered, with most candidates able to write a correct conclusion based on the information provided.
- (b) In the better responses, candidates were able to provide a safe work practice that was relevant to minimising risks in the experiment given.
- (c) In the better responses, candidates understood the need to minimise build-up of resistance by bacteria in order to maintain the effectiveness of the antibiotic in the future. Poorer responses simply restated the question.

Question 28

In the better responses, candidates demonstrated an understanding that mutations can be beneficial, by elaborating on relevant points to support their argument. Some candidates provided points for and against the quote. A significant number of candidates wrote unnecessarily long answers that exceeded the space provided.

Question 29

This question assessed skills in planning an investigation. In the better responses, candidates stated a correct hypothesis and designed an investigation that included a logical sequence of steps, an appropriate equipment list, correct identification of a dependent/independent variable, correct methods for controlling other variables, awareness of the need for repetition (or a large sample size) and risk assessment.

A significant number of candidates confused a ‘hypothesis’ with an ‘aim’, and some candidates misunderstood the phrase ‘ONE of these conditions’ and discussed only one temperature or pH. Very few candidates included risk assessment in their experimental design.

Section II – Options

Question 30 – Communication

- (a)
 - (i) A significant number of candidates were unable to name the cerebrum.
 - (ii) Better responses gave the features of and/or an annotated diagram/flowchart of the stimulus – response pathway.
- (b)
 - (i) This question required candidates to sketch in general terms the method for a first-hand investigation. Better responses clearly outlined a method using appropriate equipment.

A significant number of candidates were unable to outline a first-hand investigation to identify the relationship between wavelength, frequency and pitch.

- (ii) Few responses linked the relationship between wavelength, frequency and pitch to an increased understanding of sound or to the investigation.
- (c) Most candidates were able to identify appropriate technologies. Better responses clearly identified issues and gave points for and/or against the use of these technologies and their specific impact on society. A significant number of candidates gave general responses and did not elaborate on the specific impacts of named technologies. Many candidates wrote unnecessarily long responses to this question.
- (d)
 - (i) This part of the question was generally well answered.
 - (ii) Better responses clearly related the occurrence of colour blindness with the absorption of light by opsins.
 - (iii) Most candidates were able to describe at least two ways in which animals communicate using colour. Fewer responses were able to relate this reliance on colour to the colour vision mechanisms found in animals.

Question 31 – Biotechnology

- (a)
 - (i) Many responses included descriptions of ancient Australian Aboriginal practices without naming a biotechnology.
 - (ii) Better responses clearly distinguished early practices of biotechnology from modern methods. Few responses accurately linked selective breeding with biotechnology.
- (b)
 - (i) This question required candidates to sketch in general terms the method for a first-hand investigation. Many responses provided only a generalised method. Better responses included details of equipment and method in point form or numbered steps.
 - (ii) Few candidates were able to sketch in general terms the uses of extracted DNA.
- (c) Most candidates were able to provide examples of biotechnology and make statements about the impact of biotechnology on individuals. Better responses clearly identified issues and gave points for and/or against the use of biotechnology and its specific impact on society. Many candidates wrote unnecessarily long responses to this question.
- (d)
 - (i) Most candidates identified the basic steps involved. A significant number of candidates also named restriction and ligase enzymes.
 - (ii) Better responses clearly linked the release of the fish with disturbances to food webs or specified effects on predator populations.
 - (iii) Better responses concisely showed the relationship between a point of view and the reason.

Question 32 – Genetics: The Code Broken?

- (a) (i) Many candidates were unable to name uracil. Candidates are reminded that the letter U is not a suitable response.
- (ii) This question was generally well answered. Better responses often included clear well-labelled diagrams. Candidate responses indicate that, on the whole, transcription is better understood than translation.
- (b) (i) This question required candidates to sketch in general terms a method for modelling a process. Better responses included detail of the materials used and a step-by-step method followed. A significant number of candidates were unable to outline a method to model linkage.
- (ii) Few candidates could link their increase in understanding of linkage with the investigation.
- (c) Most candidates were able to provide examples of genetic change due to human intervention and make general statements about the impact on society. Better responses clearly identified issues and gave points for and/or against genetic change due to human intervention and its specific impact on society. Many candidates wrote unnecessarily long responses to this question.
- (d) (i) This question was well answered.
- (ii) Better responses included reasons for the differences between the graphs and some features of the mechanisms of polygenic and multiple allele inheritance.
- (iii) A significant number of candidates did not include an example in their response.

Question 33 – The Human Story

- (a) (i) Many responses included both the genus and species names without specifying the genus name.
- (ii) A significant number of responses gave features that classified prosimians as mammals rather than as primates.
- (b) (i) This question required candidates to sketch in general terms a method for modelling a process. Better responses included detail of the materials used and a step-by-step method followed. A significant number of candidates were unable to outline a method to model DNA-DNA hybridisation.
- (ii) This part of the question was well answered.
- (c) Most candidates were able to identify appropriate technologies. Better responses clearly identified issues and gave points for and/or against the use of these technologies and their specific impact on research. A significant number of candidates gave general responses and did not elaborate on the specific impacts on research. Many candidates wrote unnecessarily long responses to this question.

- (d) (i) This part of the question was well answered.
- (ii) Better responses clearly identified at least one anatomical feature from the table and explained how this had changed from a common ancestor as evidence to support human evolution.

Question 34 – Biochemistry

- (a) (i) This part of the question was well answered.
- (ii) Most responses identified the reactants and products of photosynthesis correctly. Better responses gave detailed features of the process.
- (b) (i) This question required candidates to sketch in general terms the method for a first-hand investigation. Many responses provided only a generalised method. Better responses included detail of equipment and method in point form or numbered steps.
- (ii) This part of the question was well answered.
- (c) Most candidates were able to identify new technologies. Better responses clearly identified issues and gave points for and/or against the use of these technologies and their specific impact on research. A significant number of candidates gave general responses and did not elaborate on the specific impacts on research. Many candidates wrote unnecessarily long responses to this question.
- (d) (i) Few candidates were familiar with Tswett's work or could give a correct interpretation of the results of the chromatography data.
- (ii) Better responses gave detailed characteristics and features of chlorophyll and related these to its role in the photosynthetic process.
- (iii) Better responses gave a detailed description of the role of radioactive isotopes, named specific radioactive isotopes, scientific research and the key scientists responsible for the work.

Biology

2004 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I Part A			
1	1	9.4.3 Col pt 2	H6
2	1	9.4.4 Col 1 pt 1	H6
3	1	9.4.3 Col 2 pt 3	H6
4	1	9.4.3 Col 1 pt 1	H6
5	1	9.4.4 Col 1 pt 4	H6, H14
6	1	9.2.1 Col 2 pt 9	H6
7	1	9.2.2 Col 2 pt 3	H6
8	1	9.2.3 Col 2 pt 2/Col 3 pt 1	H6
9	1	9.2.1 Col 2 pt 5	H6
10	1	9.2.1 Col 1 pt 1	H6
11	1	9.3.5 Col 2 pt 3	H9
12	1	9.3.5 pt 1	H3, H7, H10
13	1	9.3.2 Col 2 pt 4/9.1 12C	H9, H12, H14
14	1	9.3.4 pt 1	H9
15	1	9.3.3 pt 3	H6, H9
Section I Part B			
16	4	9.3.4 Col 2 pt 2	H6, H9, H13
17	7	9.3.4 Col 2 pt 1 9.3.2 Col 2 pt 1, 2, 3	H1, H2, H4, H13
18	6	9.3.1 Col 2 pt 2, 3	H10
19	4	9.4.7 Col 1 pt	H4, H13
20	3	9.4.5 Col 2, pt 1 & 2	H6
21	3	9.2.3 Col 2, pt 1, 2, 3 / Col 7, pt 4	H6
22	3	9.4.7 Col 2, pt 2	H4, H8
23	3	9.2.2 Col 2 pt 6	H6
24	3	9.2.2 Col 3, pt 5	H3
25	6	9.4.6 Col 2, pt 1 / Col 3 pt 1	H12, H13, H14
26	3	9.4.7 Col 3, pt 1	H12
27	4	9.3.4 Col 2 pt 6 9.4.3 Col 2, pt 3	H10, H14
28	3	9. 4 Col 2 pt 4	H6, H13
29	8	9.2.1 Col 2, pt 1, 2 / Col 3 pt 1	H6, H11, H12, H13

Question	Marks	Content	Syllabus outcomes
Section II			
Question 30 — Communication			
30 (a)	4	9.5.1 Col 2, pt 1 & 2	H6, H12, H13
30 (b)	6	9.5.5 & 9.5.6 Col 3 pt / Col 2, pt 3–5	H6, H12, H13
30 (c)	7	9.5	H4, H8
30 (d)	8	9.5.4 Col 1, pt 3, 4, 5 / Col 3, pt 2	H6, H7, H14, H12
Section II			
Question 31 — Biotechnology			
31 (a)	4	9.6.1 Col 3, pt 2 / Col 2, pt 2	H6, H8
31 (b)	6	9.6.5 Col 3 pt 1	H6, H11, H12, H13
31 (c)	7	9.6	H1, H3, H4, H7, H13
31 (d)	8	9.6.5 Col 2, pt 1 / Col 2 pt 2	H5, H6, H7, H14, H12
Section II			
Question 32 — Genetics: The Code Broken?			
32 (a)	4	9.7.1 Col 2, pt 1	H9
32 (b)	6	9.7.3 Col 3, pt 2 / Col 2 pt 3, 4	H3, H6, H12, H13
32 (c)	7	9.7	H1, H3, H4, H10
32 (d)	8	9.7.2 Col 2, pt 1, 2. 3. 4 / Col 3 pt 2	H9, H12, H14
Section II			
Question 33 — The Human Story			
33 (a)	4	9.8.1 Col 3, pt 1 / Col 2 pt 6	H7, H10
33 (b)	6	9.8.2 Col 2, pt 6	H3, H6, H12, H13
33 (c)	7	9.8.2	H4, H9
33 (d)	8	9.8.1, 9.8.2, 9.8.3 Col 1–5	H5, H6, H8, H9, H10, H12, H14
Section II			
Question 34 — Biochemistry			
34 (a)	4	9.9.3 Col 2 pt 3	H6
34 (b)	6	9.9.4 Col 1	H3, H6, H12, H13
34 (c)	7	9.9	H4, H9
34 (d)	8	9.9.3 Col 3, pt 3, 4	H6, H12, H14

2004 HSC Biology Marking Guidelines

Section I, Part B

Question 16 (a)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Correct nucleotide unit boxed	1

Question 16 (b)

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
Outline THREE sequenced stages of the replication process correctly	3
TWO features of the replication process outlined correctly	2
- Correct statement made relating to double-stranded nature of DNA OR - ONE correct feature relating to the replication process	1

Question 17 (a)

Outcomes assessed: H4, H13

MARKING GUIDELINES

Criteria	Marks
Correctly lists THREE possible genotypes	1

Question 17 (b)

Outcomes assessed: H13

MARKING GUIDELINES

Criteria	Marks
- Correct description of parents given as heterozygous - Demonstrates understanding that green is dominant over yellow - Indicates correctly 1560: 482 is a 3:1 ratio that shows the expected offspring from heterozygous plant OR	3
Correctly annotated Punnet square	2
Any TWO of the above	2
Any ONE of the above OR	1
Just reproducing the correct Punnet square	1

Question 17 (c)

Outcomes assessed: H1, H2

MARKING GUIDELINES

Criteria	Marks
TWO correctly described techniques using biological terms that are clearly linked to his success	3
ONE correctly described technique using biological terms that is clearly linked to his success OR	2
TWO correctly described techniques that Mendel used	2
ONE correct experimental technique used by Mendel identified	1

Question 18 (a)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
Makes a valid prediction for BOTH populations in terms of Darwin/Wallace Theory of Evolution	3
Makes a valid prediction for ONE population in terms of Darwin/Wallace Theory of Evolution	2
Makes one valid prediction without making reference to Darwin/Wallace Theory of Evolution OR	1
Provides correct information about Darwin/Wallace Theory of Evolution without making a prediction	1

Question 18 (b)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
- Clearly state that common structures support common ancestry - Demonstrate understanding of the idea of divergent evolution - Provides at least ONE correct example of vertebrate forelimbs	3
- Describes that whilst the forelimbs have the same basic arrangement, the function varies (in different organisms) - Provides at least ONE correct example of vertebrate forelimbs OR - Makes correct reference to divergent evolution and common ancestry without an example OR - Either point from above with example	2
- Identifies that different vertebrate's forelimbs have the same basic structure OR - Correctly names ONE example of vertebrate forelimbs	1

Question 19

Outcomes assessed: H4, H13

MARKING GUIDELINES

Criteria	Marks
- Describes at least TWO appropriate quarantine measures internal and/or external to Australia - Describes how they assist in protecting Australia from disease/pests, relating their effectiveness in Australia - Makes judgements about the effectiveness of quarantine measures in Australia	4
- Describes TWO appropriate quarantine measures and relates their effectiveness - Makes judgement about the effectiveness of these measures	3
- Describes an appropriate quarantine measure and relates it to the effectiveness of quarantine measure OR - Describes an appropriate quarantine measure and relates it to the effectiveness of quarantine measure - Makes a judgement about the effectiveness of a named quarantine measure	2
- Outlines an appropriate quarantine measure OR - Makes a statement about the effectiveness of quarantine in Australia	1

Question 20 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Correct identification of ONE type of T-cell	1

Question 20 (b)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
- Demonstrated understanding of the differences between B-cells and T-cells function - Relates function to role in immune system	2
Description of B-cells or T-cells function	1

Question 21

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
- Correctly explains the water balance mechanisms of fresh water fish - Clearly shows links between increased salinity and the survival of Murray Cod	3
- Describes water balance mechanism of freshwater fish - Outlines implication of increased salinity for the survival of Murray Cod	2
- Outlines water balance mechanism of fresh water fish - OR - Correct implication of increased salinity for the survival of Murray Cod	1

Question 22

Outcomes assessed: H4, H8

MARKING GUIDELINES

Criteria	Marks
Correct example used to explain how a public health strategy/pesticide/genetic engineering strategy has been used to prevent/control disease in a community	3
Correct example used to describe how a public health program/pesticide/genetic engineering strategy has been used to prevent/control disease in a community	2
- Correctly named example of public health strategy/pesticide/genetic engineering strategy used to control or prevent disease within a community - OR - Correct outline of the purpose of public health program/pesticide/genetic engineering strategy	1

Question 23

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Appropriate comparisons showing correct identification of theories relating to movement in xylem and phloem • Correct identification of material transported in each of the xylem and phloem • Correct identification of direction of movement of materials in each of xylem and phloem	3
• Any TWO of the above	2
• Any ONE of the above	1

Question 24

Outcomes assessed: H3

MARKING GUIDELINES

Criteria	Marks
• Supports arguments for continued research into development of artificial blood	3
• Arguments for the use of artificial blood, without justification OR	2
• One valid argument justified	
• States an appropriate advantage for the use of artificial blood	1

Question 25 (a)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
- Correctly labelled axes - Correct choice of graph i.e. Histogram / Bar - Key clearly indicated - Correct transposition of both sets of data - Appropriate scale – full use of grid provided (i.e. > 50% of grid)	5
- Correct choice of graph i.e. Histogram / Bar - Correctly labelled axes - Key clearly indicated Plus one of either - Correct transposition of both sets of data OR - Appropriate scale – full use of grid provided (i.e. > 50% of grid) OR - All points correct as per 5marks except 2 line graphs are drawn	4
Any THREE of these: - Correct choice of graph, ie histogram/bar - Correctly labelled axes - Key clearly indicated - Appropriate scale – full use of grid provided (i.e. > 50% of grid) - Correct transposition of both sets of data	3
- Correctly draws graph (any kind) of one or two columns with either - Appropriate key OR - Labelled axes	2
Correct attempt at representing information graphically	1

Question 25 (b)

Outcomes assessed: H14, H13

MARKING GUIDELINES

Criteria	Marks
Correct relevant data suggested	1

Question 26

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
- Correct evidence of pathogen and pest - Equipment/procedure linked is relevant to the pest/pathogen evidence provided	3
- Correct evidence of pathogen and pest - Equipment/procedure provided <u>not</u> relevant to either evidence	2
At least ONE form of correct evidence provided	1

Question 27 (a)

Outcomes assessed: H14, H10

MARKING GUIDELINES

Criteria	Marks
Correct conclusion for experiment	1

Question 27 (b)

Outcomes assessed: H14, H10

MARKING GUIDELINES

Criteria	Marks
ONE correct relevant safety practice minimising risk associated with handling or identifying microbes	1

Question 27 (c)

Outcomes assessed: H14, H10

MARKING GUIDELINES

Criteria	Marks
• Identification of the need to minimise build up of resistance by bacteria to antibiotics • Understanding the need to maintain the effectiveness of the antibiotic against that bacteria in the future	2
• Identification of the need to minimise build up of resistance by bacteria to antibiotics OR • Understanding the need to maintain the effectiveness of the antibiotic against that bacteria in the future	1

Question 28

Outcomes assessed: H6, H13

MARKING GUIDELINES

Criteria	Marks
• Appropriate issues identified and elaborated to support the argument provided • Understanding that mutations can be beneficial is demonstrated	3
• Attempt at providing appropriate argument for the issues identified OR • Understanding that mutation can be beneficial is demonstrated	2
• Correct definition of mutation OR • A qualified statement about the accuracy of the quote	1

Question 29

Outcomes assessed: H11, H12, H13, H6

MARKING GUIDELINES

Criteria	Marks
- Hypothesis stated for the factor being tested - Appropriate method very well constructed, it may include: - Logical sequence of steps - Appropriate equipment identified - Risk assessment - Identification of dependent/independent variable - Replication of large sample size - Methods of controlling other variables	7–8
- Correct hypothesis stated - Appropriate method well constructed as above, but not all of the elements of an investigation OR - Appropriate method very well constructed, as above but correct aim stated as hypothesis	5–6
- Relevant hypothesis or aim stated - Sound attempt at elements of an investigation	3–4
Basic attempt at writing a relevant report/method	2
Any relevant element of experimental method identified	1

Section II

Question 30 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Correctly identified part of brain involved in perception and interpretation of light and sound	1

Question 30 (a) (ii)

Outcomes assessed: H13, H12

MARKING GUIDELINES

Criteria	Marks
- Correctly described stimulus response pathway - May include an annotated diagram or flow chart	3
- At least TWO correct sequences of the stimulus – response pathway described OR - Correct flow chart diagram only	2
Correct identification of stimulus – response pathway with majority of terms used correctly	1

Question 30 (b) (i)

Outcomes assessed: H6, H12, H13

MARKING GUIDELINES

Criteria	Marks
Provides a thorough method that is logical, sequential and appropriate, identifying the equipment required, if relevant (this can be achieved through a fully labelled diagram)	4
Provides a method that is sequential and appropriate. (This may include a fully labelled diagram)	3
Provides a brief method that is appropriate	2
Provides some steps that could be appropriate	1

Question 30 (b) (ii)

Outcomes assessed: H6, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Relevant identification of investigation result - Link to area of investigation	2
Relevant identification of investigation result	1

Question 30 (c)

Outcomes assessed: H4, H8

MARKING GUIDELINES

Criteria	Marks
- Appropriate issues identified with relevant points for and/or against - Well developed argument demonstrating (link) understanding of impact on society using relevant examples (Answer may be represented as diagram, table, annotated flowcharts, mind maps) - Argument follows a logical sequence, with the use of appropriate biological terminology	6–7
- Appropriate issues identified with an attempt to illustrate points for and against - Simply identified link with impact on society with at least ONE example - Limited use of biological terminology	4–5
- Appropriate issues identified - Attempt at demonstrating an understanding of the impact on society	2–3
Identified an appropriate issue relevant to the theme	1

Question 30 (d) (i)

Outcomes assessed: H6, H7, H14, H12

MARKING GUIDELINES

Criteria	Marks
• Correct identification of THREE types of cones, red, blue and green • Understanding that variance in stimulation of the THREE cones results in combination of different colours	2
• Correct identification of THREE types of cones, red, blue and green OR • Understanding that variance in stimulation of the THREE cones results in combination of different colours	1

Question 30 (d) (ii)

Outcomes assessed: H6, H7

MARKING GUIDELINES

Criteria	Marks
• Correct identification of cause for colour blindness being missing or insufficient particular opsins in the respective colour blindness • Link with inability to detect certain colours or contrast	2
• Correct identification of cause for colour blindness for missing or insufficient particular opsins • No link	1

Question 30 (d) (iii)

Outcomes assessed: H6, H7

MARKING GUIDELINES

Criteria	Marks
- Correctly describes the uses of colour for communication in animals - Identifies at least TWO different animals and describe how they perceive colour	4
- Correctly outlines the uses of colour for communication in animals - Identify at least TWO different methods animals may detect colour	3
- Correct statement about the use of colour for communication in animals AND/EITHER - At least ONE method of colour vision in an identified animal OR - Colour vision clearly linked to animal's environment	2
- ONE example of use of colour for communication in an animal OR - Correct statement about the purpose of colour for communication in animals	1

Question 31 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
- Name of an appropriate Australian Aboriginal use of biotechnology OR - A clear description of aquaculture	1

Question 31 (a) (ii)

Outcomes assessed: H6, H8

MARKING GUIDELINES

Criteria	Marks
Provides correct characteristics and features of early biotechnology and describes how breeding animals is an example	3
- Correct description of early biotechnology OR - Defines what is meant by early biotechnology	2
- Makes a statement about the use of biotechnology OR - Defines biotechnology OR - Identifies an example of biotechnology	1

Question 31 (b) (i)

Outcomes assessed: H6, H12, H13

MARKING GUIDELINES

Criteria	Marks
Provides a thorough method that is logical, sequential and appropriate, identifying the equipment required, if relevant (this can be achieved through a fully labelled diagram)	4
Provides a method that is sequential and appropriate. (This may include a fully labelled diagram)	3
Provides a brief method that is appropriate	2
Provides some steps that could be appropriate	1

Question 31 (b) (ii)

Outcomes assessed: H6, H12, H13

MARKING GUIDELINES

Criteria	Marks
Correctly outlines TWO uses of extracted DNA in Biotechnology	2
- Correctly outlines ONE use of extracted DNA OR - Identifies TWO uses	1

Question 31 (c)

Outcomes assessed: H1, H3, H4, H7, H13

MARKING GUIDELINES

Criteria	Marks
- Appropriate issues identified with relevant points for and/or against - Well developed argument demonstrating a link between understanding of impact on society using relevant examples - Argument follows a logical sequence with the use of appropriate biological vocabulary	6–7
- Appropriate issues identified with an attempt to illustrate points for and/or against - Simple identified links with impact on society with at least ONE example - Limited use of biological vocabulary	4–5
- Appropriate issue(s) identified - Attempt at demonstrating an understanding of the impact on society	2–3
Identifies an appropriate issue relevant to the theme	1

Question 31 (d) (i)

Outcomes assessed: H5, H6, H7, H12, H14

MARKING GUIDELINES

Criteria	Marks
Correctly outlines in sequence the process of gene splicing to produce recombinant DNA	2
- Outlines ONE correct step OR - States TWO correct steps OR - Makes a correct statement about production of recombinant DNA	1

Question 31 (d) (ii)

Outcomes assessed: H5, H6, H7, H12, H14

MARKING GUIDELINES

Criteria	Marks
Puts forward a viable proposal as to the effect on the environment	2
- Makes a statement about the consequence of mating fluorescing and non fluorescing fish OR - Makes a statement about the effect on the environment	1

Question 31 (d) (iii)

Outcomes assessed: H5, H6, H7, H12, H14

MARKING GUIDELINES

Criteria	Marks
• Correctly explains TWO differing points of view	4
• Identifies differing points of view • Provides an explanation for ONE point of view	3
• Outlines ONE point of view OR • Provides an explanation about the use of biotechnology for producing zebra fish	2
• Any correct statement or example of biotechnology	1

Question 32 (a) (i)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies N base	1

Question 32 (a) (ii)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• Coherent answer, including both transcription and translation • Use of correct terms with logical sequence of steps leading to polypeptide production	3
• Translation outlined and transcriptions outlined	2
• Transcription outlined <u>or</u> translation outlined	1

Question 32 (b) (i)

Outcomes assessed: H3, H6, H12, H13

MARKING GUIDELINES

Criteria	Marks
Provides a thorough method that is logical, sequential and appropriate, identifying the equipment required, if relevant (this can be achieved through a fully labelled diagram)	4
Provides a method that is sequential and appropriate. (This may include a fully labelled diagram)	3
Provides a brief method that is appropriate	2
Provides some steps that could be appropriate	1

Question 32 (b) (ii)

Outcomes assessed: H3, H6, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Relevant identification of investigation result - Link to area of investigation	2
Relevant identification of investigation result	1

Question 32 (c)

Outcomes assessed: H4, H3, H1, H10

MARKING GUIDELINES

Criteria	Marks
- Appropriate issues identified with relevant points for and/or against - Well developed argument demonstrating (link) understanding of impact on society using relevant examples - Argument follows a logical sequence with the use of appropriate biological vocabulary	6–7
- Appropriate issues identified with an attempt to illustrate points for and against - Simply identified link with impact on society with at least ONE example - Limited use of biological vocabulary	4–5
- Appropriate issues identified - Attempt at demonstrating an understanding on the impact on society	2–3
Identified an appropriate issue relevant to the theme	1

Question 32 (d) (i)

Outcomes assessed: H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
Correctly identifies egg size as a polygenic trait	1

Question 32 (d) (ii)

Outcomes assessed: H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
- A <u>clear</u> discussion of the reasons for the differences in the kind of shapes of the graphs/info contained/illustrated - Fully describes mechanisms of both polygenic and multiple allele inheritance	4
- Fully discusses mechanisms of both polygenic and multiple allele inheritance OR - Describes the two mechanisms including reference to shape or kind of graph	3
- Describes mechanisms of both polygenic and multiple allele inheritance OR - Describes ONE mechanism and identifies reason for kind or shape of graphs it relates to OR - Fully describes one mechanism	2
- Describes mechanism of polygenic or multiple allele inheritance OR - Relates each mechanism to the kind or shape of the corresponding graph OR - Superficial comparison between polygenic and multiple allele inheritance	1

Question 32 (d) (iii)

Outcomes assessed: H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
• Outlines giving a clear understanding of the use of highly variable DNA • An example/named use	3
• Outlines giving clear understanding of the use of highly variable DNA OR • Outline the use of highly variable DNA, giving example	2
• Indicates the basis of the use of highly variable DNA OR • ONE example e.g. Pedigree testing/paternity testing	1

Question 33 (a) (i)

Outcomes assessed: H7, H10

MARKING GUIDELINES

Criteria	Marks
• Correctly names the genus for humans	1

Question 33 (a) (ii)

Outcomes assessed: H7, H10

MARKING GUIDELINES

Criteria	Marks
• Correctly names and describes THREE or more characteristics of prosimians	3
• Correctly names TWO characteristics and describes these two OR • Names THREE characteristics and describes at least ONE	2
• Names and describes ONE characteristic OR • Names at least TWO characteristics	1

Question 33 (b) (i)

Outcomes assessed: H3, H13, H12 & H6

MARKING GUIDELINES

Criteria	Marks
Provides a thorough method that is logical, sequential and appropriate, identifying the equipment required, if relevant (this can be achieved through a fully labelled diagram)	4
Provides a method that is sequential and appropriate. (This may include a fully labelled diagram)	3
Provides a brief method that is appropriate	2
Provides some steps that could be appropriate	1

Question 33 (b) (ii)

Outcomes assessed: H3, H1

MARKING GUIDELINES

Criteria	Marks
Relevant identification of investigation result	2
Link to area of investigation	
Relevant identification of investigation result	1

Question 33 (c)

Outcomes assessed: H4, H9

MARKING GUIDELINES

Criteria	Marks
- Appropriate technologies identified with relevant points for and/or against - Well developed argument demonstrating (link) understanding of impact on research using relevant examples - Argument follows a logical sequence, with the use of appropriate biological vocabulary	6–7
- Appropriate technologies identified with an attempt to illustrate points for and against - Simply identified link with impact on research with at least ONE example - Limited use of biological vocabulary	4–5
- Appropriate technologies identified or technology outlined - Attempt at demonstrating an understanding of the impact on research	2–3
Identified an appropriate technique relevant to the theme	1

Question 33 (d) (i)

Outcomes assessed: H5, H8, H10, H9, H6, H14, H12

MARKING GUIDELINES

Criteria	Marks
- Correctly draws and labels table - Correctly completes all <u>three</u> rows (6 cells) of the table	4
- Draws and labels a table - Correctly completes at least TWO complete rows (4 cells) of the table	3
- Draws and labels a table - Correctly completes at least ONE row (2 cells) of the table	2
- Draw table – three rows/three columns – correct label on each column/row OR - Correct description of difference of at least one feature without a table	1

Question 33 (d) (ii)

Outcomes assessed: H10, H8, H5, H14, H12

MARKING GUIDELINES

Criteria	Marks
• Identifies TWO anatomical features from their table and explains evolutionary perspective	4
• Identifies TWO anatomical features from the table and explains one from an evolution perspective	3
• Identifies ONE anatomical feature and explains one evolutionary perspective	2
• Identifies at least ONE anatomical feature	1

Question 34 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Correct identification of organelle	1

Question 34 (a) (ii)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Correct description of photosynthesis indicating products, reactants and energy transfer	3
• Outlines some aspects correctly	2
• Identifies some aspects correctly	1

Question 34 (b) (i)

Outcomes assessed: H3, H6, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Provides a thorough method that is logical, sequential and appropriate, identifying the equipment required, if relevant (this can be achieved through a fully labelled diagram)	4
• Provides a method that is sequential and appropriate. (This may include a fully labelled diagram)	3
• Provides a brief method that is appropriate	2
• Provides some steps that could be appropriate	1

Question 34 (b) (ii)

Outcomes assessed: H3, H6, H12

MARKING GUIDELINES

Criteria	Marks
- Relevant identification of investigation result - Link to area of investigation	2
Relevant identification of investigation result	1

Question 34 (c)

Outcomes assessed: H4, H9

MARKING GUIDELINES

Criteria	Marks
- Appropriate technologies identified with relevant points for and/or against - Well developed argument demonstrating (link) understanding of impact on research using relevant examples - Argument follows a logical sequence, with the use of appropriate biological vocabulary	6–7
- Appropriate technologies identified with an attempt to illustrate points for and against - Simply identified link with impact on research with at least ONE example - Limited use of biological vocabulary	4–5
- Appropriate technologies identified or technology outlined - Attempt at demonstrating an understanding of the impact on research	2–3
Identified an appropriate technique relevant to the theme	1

Question 34 (d) (i)

Outcomes assessed: H6, H14, H12

MARKING GUIDELINES

Criteria	Marks
- Chromatography AND - Green plants have more than one pigment (separates by chromatography)	2
ONE of the above	1

Question 34 (d) (ii)

Outcomes assessed: H6, H14, H12

MARKING GUIDELINES

Criteria	Marks
Any TWO of these described - Energized chlorophyll produces electrons - Electrons used to produce H and O from H₂O, ATP and NADPH₂ - H + O components are used to produce glucose OR - Any ONE thoroughly explained	2
ONE of the above	1

Question 34 (d) (iii)

Outcomes assessed: H6, H14, H12

MARKING GUIDELINES

Criteria	Marks
- Thorough explanation of the use of radioactive isotopes to describe photosynthetic pathways AND - Names Hill or Calvin or Ruben	4
- Correct explanation of the use of radioactive isotopes to describe photosynthetic pathways AND - Names Hill or Calvin or Ruben	3
Correct explanation of the use of radioactive isotopes to describe photosynthetic pathways	2
- Correctly states the role of isotopes OR - Names scientist correctly	1